

corrigé zakito

Énoncé

Autocorrection A

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}, \quad \text{(i) } AB = 40I_4, \quad \text{(ii) } AB = \text{diag}(2, 4, -2).$$

Exercice 1

$$v^T v = \sum_{i=1}^n v_i^2 = 0 \iff \forall i, v_i = 0 \iff v = 0.$$

Exercice 2

$$(JMJ)_{ij} = \sum_{k,\ell=1}^n m_{k\ell} \Rightarrow JMJ = \left(\sum_{k,\ell=1}^n m_{k\ell} \right) J.$$

Exercice 3

$$E_{ij}E_{k\ell} = \delta_{jk}E_{i\ell}.$$

$$\text{(i) } A = E_{11}, \quad B = E_{21} \Rightarrow AB = 0, \quad BA = E_{21} \neq 0.$$

$$\text{(ii) } A = E_{11}, \quad B = E_{22} \Rightarrow AB = BA = 0, \quad A \neq 0, \quad B \neq 0.$$

Exercice 4

1

$$S(a, b) + S(c, d) = S(a + c, b + d),$$

$$S(a, b)S(c, d) = S(ac - bd, ad + bc) = S(c, d)S(a, b).$$

2

$$S(a, b)^T = S(a, -b), \quad S(a, b)S(a, b)^T = (a^2 + b^2)I_2.$$

3

$$(a, b) \neq (0, 0) \Rightarrow S(a, b)^{-1} = \frac{1}{a^2 + b^2} S(a, -b).$$

$$S(0, 0) = 0 \Rightarrow S(a, b) \in GL_2(\mathbb{R}) \iff a^2 + b^2 > 0.$$

4

$$S(\cos \alpha, \sin \alpha)S(\cos \beta, \sin \beta) = S(\cos(\alpha + \beta), \sin(\alpha + \beta)).$$

$$S(\cos \theta, \sin \theta)^n = S(\cos(n\theta), \sin(n\theta)) \quad (n \in \mathbb{Z}).$$

Exercice 5

$$Q(a, b) = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}.$$

1

$$Q(a, b) + Q(c, d) = Q(a + c, b + d),$$

$$Q(a, b)Q(c, d) = Q(ac - b\bar{d}, ad + b\bar{c}).$$

2

$$Q(a, b)Q(\bar{a}, -b) = (|a|^2 + |b|^2)I_2.$$

$$\mathbb{H} \cap GL_2(\mathbb{C}) = \mathbb{H} \setminus \{0\}, \quad Q(a, b)^{-1} = \frac{Q(\bar{a}, -b)}{|a|^2 + |b|^2} \in \mathbb{H}.$$

Exercice 6

$$X^2 + X = J \Rightarrow XJ = JX.$$

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad XJ = JX \Rightarrow c = b, \quad d = a.$$

$$X^2 + X = J \iff a^2 + b^2 + a = 1, \quad b(2a + 1) = 1.$$

$$a, b \in \mathbb{Z} \Rightarrow (b, 2a + 1) = (1, 1) \text{ ou } (-1, -1).$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{ou} \quad X = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}.$$

Exercice 7

$$M = \underbrace{\frac{M + M^T}{2}}_S + \underbrace{\frac{M - M^T}{2}}_A, \quad S^T = S, \quad A^T = -A.$$

$$M = S_1 + A_1 \Rightarrow M^T = S_1 - A_1 \Rightarrow S_1 = S, \quad A_1 = A.$$

Exercice 8

$$\text{tr}(PQ) = \sum_{i,j} p_{ij}q_{ji} = \text{tr}(QP).$$

$$PQ - QP = I_n \Rightarrow 0 = \text{tr}(PQ - QP) = n, \quad \perp \quad (\mathbb{K} = \mathbb{R} \text{ ou } \mathbb{C}).$$

Exercice 9

$$(AB)^T = B^T A^T = BA \Rightarrow AB \in S_n(\mathbb{R}) \iff AB = BA.$$

Exercice 10

$$(AD - DA)_{ij} = (d_j - d_i)a_{ij}.$$

$$AD = DA \iff \forall i \neq j, a_{ij} = 0 \iff A \text{ diagonale.}$$

Exercice 11

$$\text{Dimension corrigée : } A \in M_{n+1}(\mathbb{K}).$$

$$AD = DA \iff (d_j - d_i)a_{ij} = 0.$$

$$A = \text{diag}(a_1, \dots, a_{n-1}, B), \quad a_i \in \mathbb{K}, \quad B \in M_2(\mathbb{K}).$$

Exercice 12

$$AE_{kk} = E_{kk}A \quad (1 \leq k \leq n) \Rightarrow A = \text{diag}(a_1, \dots, a_n).$$

$$AE_{ij} = a_i E_{ij}, \quad E_{ij}A = a_j E_{ij} \Rightarrow a_i = a_j.$$

$$A = \lambda I_n.$$

Exercice 13

$$AA^T = A^T A \Rightarrow \sum_{j=1}^n a_{1j}^2 = \sum_{j=1}^n a_{j1}^2 = a_{11}^2 \Rightarrow a_{1j} = 0 \quad (j > 1).$$

$$A = \text{diag}(a_{11}, B), \quad B \text{ triangulaire supérieure}, \quad BB^T = B^T B.$$

Récurrence sur n : B , puis A , est diagonale.

Exercice 14

$$n = 1 : \quad \mathcal{C} = M_1(\mathbb{K}).$$

$$n = 2 : \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad AJ = JA \iff A = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

Pour $n \geq 3$, soit $F_{ij} = E_{ij} - E_{ji}$.

$$AF_{ij} = F_{ij}A \Rightarrow AF_{ij}^2 = F_{ij}^2A, \quad F_{ij}^2 = -E_{ii} - E_{jj}.$$

Pour i, j, k distincts,

$$2E_{ii} = -(F_{ij}^2 + F_{ik}^2 - F_{jk}^2) \Rightarrow AE_{ii} = E_{ii}A.$$

$$A = \text{diag}(a_i), \quad AF_{ij} = F_{ij}A \Rightarrow a_i = a_j.$$

$$\mathcal{C} = \mathbb{K}I_n \quad (n \geq 3).$$

Exercice 15

$$AB = A + B \Rightarrow (A - I)(B - I) = I.$$

$$A - I \text{ et } B - I \text{ inversibles} \Rightarrow (B - I)(A - I) = I \Rightarrow BA = A + B = AB.$$

Autocorrection B

$$s = \sin x, \quad c = \cos x, \quad A^2 = \begin{pmatrix} -c^2 & -sc & c \\ -sc & -s^2 & s \\ -c & -s & 1 \end{pmatrix}, \quad A^3 = 0.$$

$$(I_3 + A)^m = I_3 + mA + \frac{m(m-1)}{2}A^2 \quad (m \in \mathbb{Z}).$$

Pour $m \geq 0$, binôme de Newton ; pour $m < 0$, $(I + A)^{-1} = I - A + A^2$ et récurrence descendante.

Exercice 16

L'exposant est noté $m \geq 1$.

(i)

$$(aI_n)^m = a^m I_n.$$

(ii)

$$A^2 = 2A \Rightarrow A^m = 2^{m-1} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

(iii)

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}^m = \begin{pmatrix} 1 & \sum_{j=0}^{m-1} 2^j \\ 0 & 2^m \end{pmatrix} = \begin{pmatrix} 1 & 2^m - 1 \\ 0 & 2^m \end{pmatrix}.$$

(iv)

$$P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad Q = I - P, \quad P^2 = P, \quad Q^2 = Q, \quad PQ = QP = 0.$$

$$A = 3P - Q \Rightarrow A^m = \frac{1}{2} \begin{pmatrix} 3^m + (-1)^m & 3^m - (-1)^m \\ 3^m - (-1)^m & 3^m + (-1)^m \end{pmatrix}.$$

(v)

$$A = aI + bE_{12}, \quad E_{12}^2 = 0 \Rightarrow A^m = a^m I + ma^{m-1}bE_{12}.$$

Pour $m = 1$, $A^1 = A$; la formule vaut donc aussi lorsque $a = 0$.

(vi)

$$A^m = \begin{pmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{pmatrix}.$$

(vii)

$$A^2 = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \quad A^3 = 3A.$$

$$A^{2r+1} = 3^r A \quad (r \geq 0), \quad A^{2r} = 3^{r-1} A^2 \quad (r \geq 1).$$

(viii)

$$A = J_3 - I_3, \quad P = J_3/3, \quad P^2 = P.$$

$$A^m = 2^m P + (-1)^m (I_3 - P) = (-1)^m I_3 + \frac{2^m - (-1)^m}{3} J_3.$$

(ix)

$$P = J_n/n, \quad Q = I_n - P, \quad A = (a + (n-1)b)P + (a-b)Q.$$

$$A^m = (a-b)^m I_n + \frac{(a + (n-1)b)^m - (a-b)^m}{n} J_n.$$

(x)

$$J_n^2 = nJ_n \Rightarrow J_n^m = n^{m-1}J_n.$$

Exercice 17

$$\lambda = LC = \sum_{j=1}^n \ell_j c_j \in \mathbb{C}, \quad (LC)^p = \lambda^p.$$

$$(CL)^2 = C(LC)L = \lambda CL \Rightarrow (CL)^p = \lambda^{p-1} CL \quad (p \geq 1).$$

$$(LC)^0 = 1, \quad (CL)^0 = I_n.$$

Exercice 18

1

$$Ae_1 = e_2, \quad Ae_2 = e_3, \quad Ae_3 = e_4, \quad Ae_4 = xe_1 \Rightarrow A^4 = xI_4.$$

2

$$x = 0 \Rightarrow Ae_4 = 0 \Rightarrow A \text{ non inversible.}$$

$$x \neq 0 \Rightarrow A^{-1} = x^{-1}A^3.$$

3

$$n = 4q + r, \quad q \in \mathbb{Z}, \quad r \in \{0, 1, 2, 3\} \Rightarrow A^n = x^q A^r.$$

$$A^2 = \begin{pmatrix} 0 & 0 & x & 0 \\ 0 & 0 & 0 & x \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & x & 0 & 0 \\ 0 & 0 & x & 0 \\ 0 & 0 & 0 & x \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Exercice 19

1

Le coefficient $(2, 3)$ imprimé est xz . Pour $(x, y, z) = (1, 0, 1)$,

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \text{rg } A = 2 > 1.$$

La factorisation demandée est donc impossible en général. Avec $a_{23} = yz$,

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \tilde{A} = vv^T.$$

2

$$s = x^2 + y^2 + z^2, \quad \tilde{A}^m = s^{m-1} \tilde{A} \quad (m \geq 1), \quad \tilde{A}^0 = I_3.$$

Pour la matrice imprimée, $d = z(x - y)$:

$$A = UV, \quad U = \begin{pmatrix} x & 0 \\ y & 1 \\ z & 0 \end{pmatrix}, \quad V = \begin{pmatrix} x & y & z \\ 0 & 0 & d \end{pmatrix}, \quad H = VU = \begin{pmatrix} s & y \\ dz & 0 \end{pmatrix}.$$

$$A^m = UH^{m-1}V \quad (m \geq 1).$$

$$c = dyz, \quad H^2 = sH + cI_2, \quad F_0 = 0, \quad F_1 = 1, \quad F_{r+1} = sF_r + cF_{r-1}.$$

$$H^r = F_r H + cF_{r-1}I_2 \quad (r \geq 1), \quad F_r = \sum_{j=0}^{\lfloor (r-1)/2 \rfloor} \binom{r-1-j}{j} s^{r-1-2j} c^j \quad (r \geq 1).$$

Exercice 20

$$B = \begin{pmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix}, \quad B^2 = I_2, \quad P_{\pm} = \frac{I_2 \pm B}{2}.$$

$$P_{\pm}^2 = P_{\pm}, \quad P_+ P_- = 0, \quad A = xI_2 + B = (x+1)P_+ + (x-1)P_-.$$

$$A^m = \frac{(x+1)^m + (x-1)^m}{2} I_2 + \frac{(x+1)^m - (x-1)^m}{2} B \quad (m \geq 0).$$

Si $x \neq \pm 1$, la formule vaut aussi pour $m \in \mathbb{Z}$.

Exercice 21

1

$$A = B + C, \quad A^2 = B + 2C, \quad A^3 = B + 3C \Rightarrow A^3 - 2A^2 + A = 0.$$

$$A^{r+2} - 2A^{r+1} + A^r = 0 \quad (r \geq 1).$$

$$A^{r+1} - A^r = A^2 - A = C \Rightarrow A^r = A + (r-1)C = B + rC.$$

2

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A = B + C.$$

$$B^2 = B, \quad BC = CB = C, \quad C^2 = 0 \Rightarrow A^r = B + rC \quad (r \geq 1).$$

Exercice 22

$$A^p = 0, \quad B^q = 0, \quad AB = BA.$$

$$(A + B)^{p+q-1} = \sum_{k=0}^{p+q-1} \binom{p+q-1}{k} A^k B^{p+q-1-k} = 0,$$

car $k \geq p$ ou $p+q-1-k \geq q$.

$$(AB)^p = A^p B^p = 0.$$

Sans commutation :

$$A = E_{12}, \quad B = E_{21}, \quad A^2 = B^2 = 0, \quad (A + B)^2 = I_2, \quad (AB)^m = E_{11} \neq 0.$$

Exercice 23

Pour T strictement triangulaire supérieure,

$$(T^r)_{ij} = \sum_{i=i_0 < i_1 < \dots < i_r=j} t_{i_0 i_1} \cdots t_{i_{r-1} i_r}.$$

$$r = n \Rightarrow \{1 \leq i_0 < \dots < i_n \leq n\} = \emptyset \Rightarrow T^n = 0.$$

Le cas inférieur se déduit par transposition.

Exercice 24

$$A = J_3 - I_3, \quad J_3^2 = 3J_3 \Rightarrow A^2 = J_3 + I_3 = A + 2I_3.$$

$$A \frac{A - I_3}{2} = I_3 \Rightarrow A^{-1} = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

Exercice 25

$$N = \sum_{i=1}^{n-1} E_{i,i+1}, \quad N^n = 0, \quad A = I + N + \dots + N^{n-1}.$$

$$(I - N)A = A(I - N) = I \Rightarrow A^{-1} = I - \sum_{i=1}^{n-1} E_{i,i+1}.$$

Exercice 26

1

$$M = (a - b)I + bJ, \quad J^2 = nJ, \quad \alpha = a - b, \quad \beta = a + (n - 1)b.$$

$$M^2 = (\alpha + \beta)M - \alpha\beta I = (2a + (n - 2)b)M - (a - b)(a + (n - 1)b)I.$$

2

$$M\mathbf{1} = \beta\mathbf{1}, \quad M(e_1 - e_2) = \alpha(e_1 - e_2).$$

$$\alpha\beta = 0 \Rightarrow \ker M \neq \{0\}.$$

$$\alpha\beta \neq 0 \Rightarrow M^{-1} = \frac{(\alpha + \beta)I - M}{\alpha\beta} = \frac{1}{a - b}I - \frac{b}{(a - b)(a + (n - 1)b)}J.$$

Exercice 27

$$(F\overline{F})_{ij} = \sum_{k=0}^{n-1} \omega^{(i-j)k} = \begin{cases} n & i = j, \\ \frac{1 - \omega^{(i-j)n}}{1 - \omega^{i-j}} = 0 & i \neq j. \end{cases}$$

$$F\overline{F} = nI_n, \quad F^{-1} = \frac{1}{n}\overline{F}.$$

Exercice 28

1

Si la colonne j de A est nulle, alors, pour toute B , $(BA)_{ij} = \sum_k b_{ik}a_{kj} = 0$. Ainsi $BA \neq I_n$.
 $AE_{jj} = 0 = A0$, $E_{jj} \neq 0 \Rightarrow A$ non simplifiable à gauche $\Rightarrow A$ non inversible.

2

Si la ligne i de A est nulle, alors, pour toute B , $(AB)_{ij} = \sum_k a_{ik}b_{kj} = 0$. Ainsi $AB \neq I_n$.
 $E_{ii}A = 0 = 0A$, $E_{ii} \neq 0 \Rightarrow A$ non simplifiable à droite.
 A^T possède une colonne nulle $\Rightarrow A^T$ non inversible $\Rightarrow A$ non inversible.

Exercice 29

1

$$M \left(-\frac{1}{a_0} \sum_{k=1}^d a_k M^{k-1} \right) = I_n.$$

$$M^{-1} = -\frac{1}{a_0} \sum_{k=1}^d a_k M^{k-1}.$$

2

$$M = J_n - I_n, \quad M^2 = (n-2)M + (n-1)I_n.$$

$$n \geq 2 \Rightarrow M^{-1} = \frac{M - (n-2)I_n}{n-1} = -I_n + \frac{1}{n-1}J_n.$$

Exercice 30

1

$$(I - M)(I + M + \dots + M^{p-1}) = I - M^p = I.$$

$$(I - M)^{-1} = \sum_{k=0}^{p-1} M^k.$$

2

$$M = I - A = \begin{pmatrix} 0 & -3 & 1 \\ -3 & 0 & 2 \\ 3 & -6 & 0 \end{pmatrix}, \quad M^2 = \begin{pmatrix} 12 & -6 & -6 \\ 6 & -3 & -3 \\ 18 & -9 & -9 \end{pmatrix}, \quad M^3 = 0.$$

$$A^{-1} = I + M + M^2 = \begin{pmatrix} 13 & -9 & -5 \\ 3 & -2 & -1 \\ 21 & -15 & -8 \end{pmatrix}.$$

Exercice 31

$$A = L + D + U, \quad D = \text{diag}(a_{ii}).$$

L strictement triangulaire inférieure, U strictement triangulaire supérieure.

$$\lambda \notin \{0, a_{11}, \dots, a_{nn}\}, \quad A = \underbrace{(U + \lambda I)}_B + \underbrace{(L + D - \lambda I)}_C.$$

B, C triangulaires à diagonales non nulles $\Rightarrow B, C \in GL_n(\mathbb{K})$.